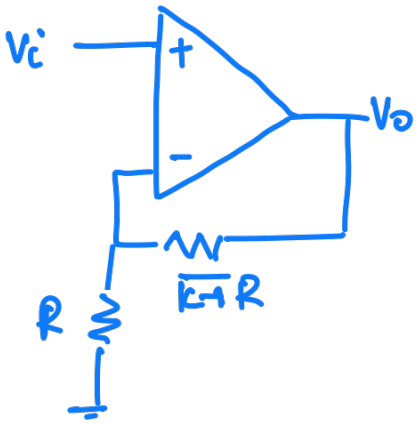
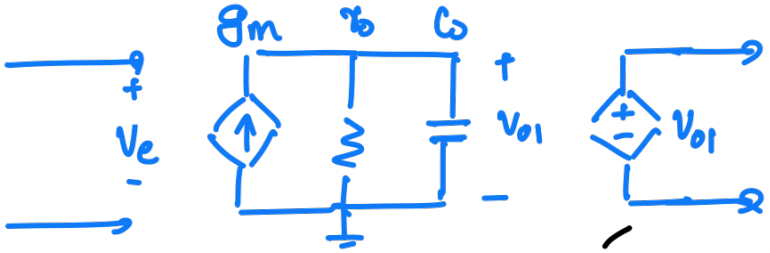
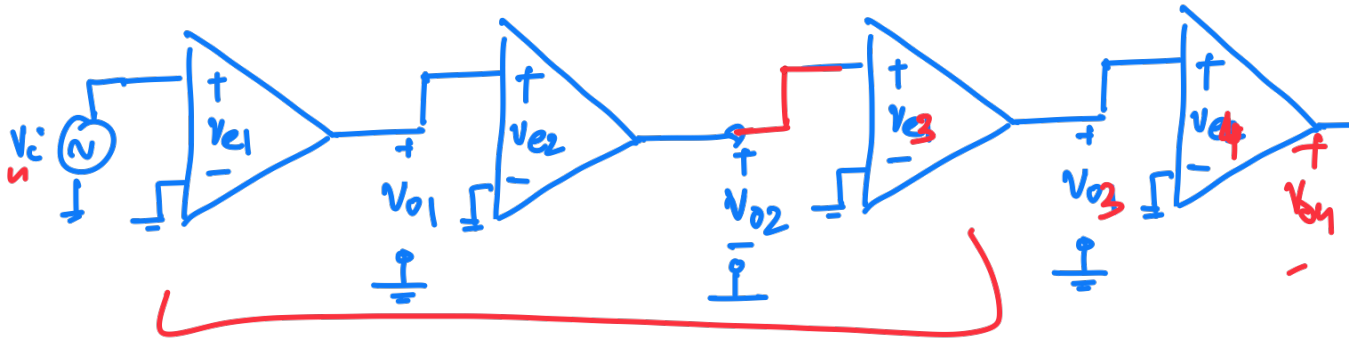
Lecture # 11





$$\frac{V_o(s)}{V_i(s)} = \frac{k}{1+\frac{k}{A(s)}} \quad ; \quad A(s) = \frac{A_0}{1+s/\omega_p} = \frac{g_m r_o}{1+sC_o r_o}$$

- ϵ_s
- ϵ_d

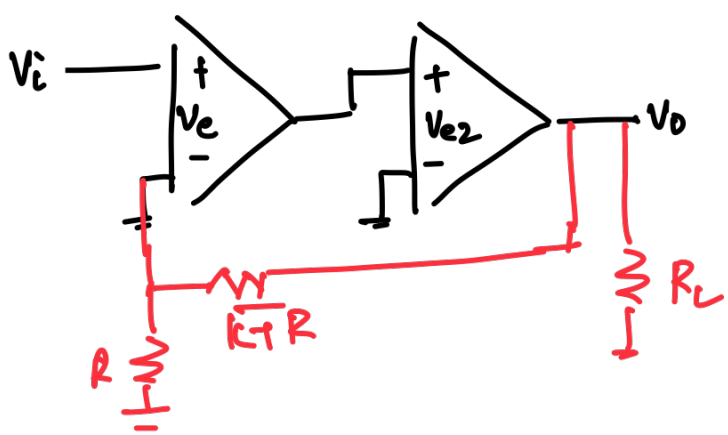


$$\frac{V_{o2}}{V_i} = \left(\frac{A_0}{1+s/\omega_p}\right)^2$$

$$\frac{V_{o4}}{V_i} = \left(\frac{A_0}{1+s/\omega_p}\right)^4$$

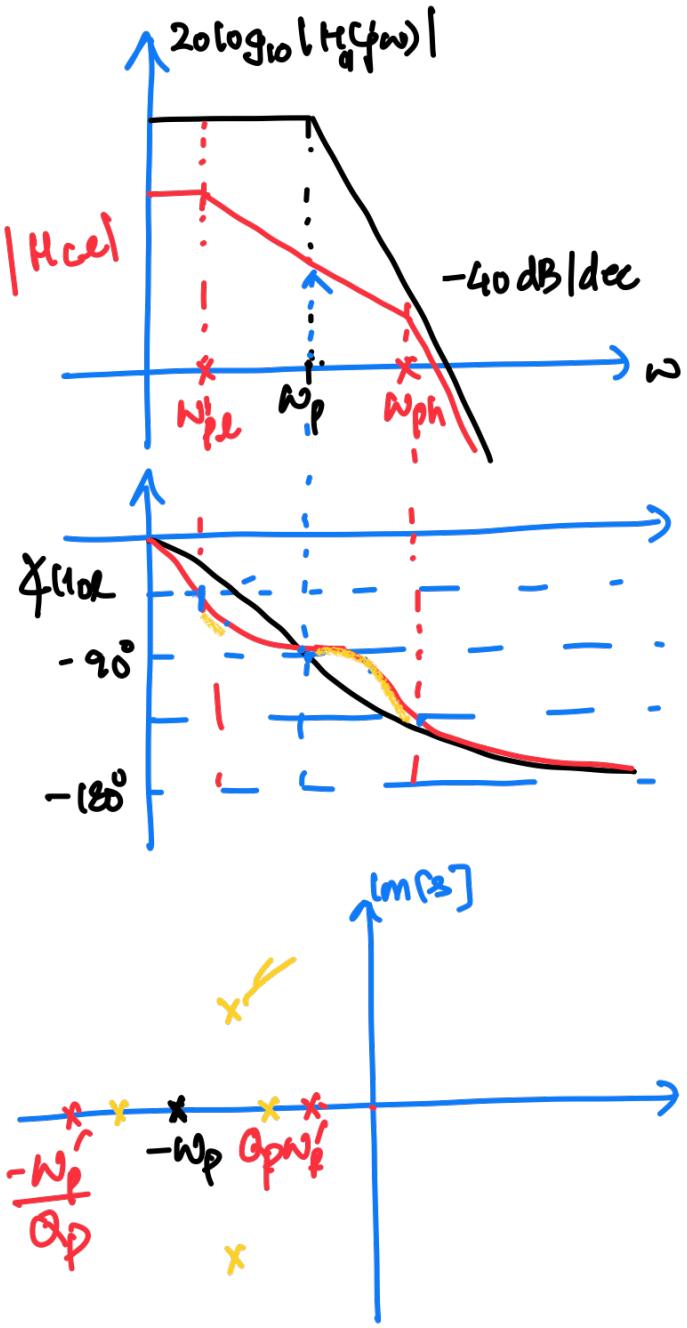
$$\sin(\omega_{in}t+\phi)$$

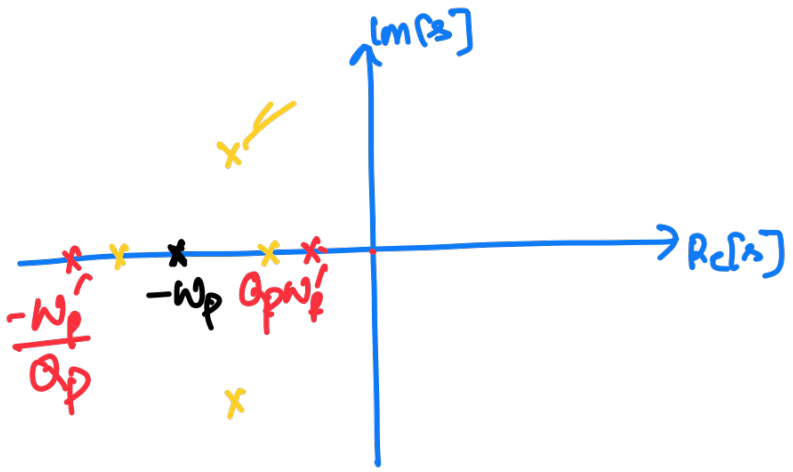
$$\sin(\omega_{in}t+2\pi+\phi)$$



$$H_{ol} = \frac{V_o}{V_e} = \frac{A_0^2}{(1+s/\omega_p)^2}$$

$$H_{OL} = \frac{A_0^2}{(1 + s/\omega_p)^2}$$





$$\frac{V_o}{V_i} = \frac{k}{1 + \frac{k}{H_{OL}}} = H_{cl}(s)$$

$$= \frac{k}{1 + \frac{k(1 + s/\omega_p)^2}{A_{02}}} \quad ; A_{02} = A_0^2$$

$$= \frac{k}{1 + \frac{k}{A_{02}}\left(1 + \frac{s^2}{\omega_p^2} + \frac{2s}{\omega_p}\right)}$$

$$= \frac{k}{1 + \frac{k}{A_{02}} + \frac{k}{A_{02}}\frac{2}{\omega_p}s + \frac{k}{A_{02}}\frac{s^2}{\omega_p^2}}$$

$$= \underbrace{\frac{k}{1 + k/A_{02}}} \times \frac{1}{1 + \frac{2s/\omega_p}{\left(\frac{A_{02}}{k} + 1\right)} + \frac{s^2/\omega_p^2}{\left(\frac{A_{02}}{k} + 1\right)}}$$

$$H_{cl}(s) = \frac{k}{1 + k/A_{02}} \quad \underbrace{\frac{1}{1 + \frac{2}{\left(\frac{A_{02}}{k} + 1\right)}\frac{s}{\omega_p} + \frac{1}{\left(\frac{A_{02}}{k} + 1\right)}\frac{s^2}{\omega_p^2}}}$$

$$= \frac{k}{1 + k/A_{02}} \quad \underbrace{\frac{1}{1 + \frac{s}{\omega_p' Q_p} + \frac{s^2}{\omega_p'^2}}}$$

$$\omega_p' = \omega_p \sqrt{\frac{A_{02}}{K} + 1}$$

$$\omega_p' Q_p = \frac{\omega_p}{2}\left(\frac{A_{02}}{K} + 1\right)$$

$$Q_p = \frac{1}{2}\sqrt{\frac{A_{02}}{K} + 1}$$ Quality factor.

$$D(s) = \frac{s^2}{\omega_p'^2} + \frac{s}{\omega_p' Q_p} + 1$$

$$D(s) = 0$$

$$\frac{s}{\omega_p'} = \frac{-\frac{1}{Q_p} \pm \sqrt{\frac{1}{Q_p^2} - 4}}{2}$$

$$s = -\omega_p' \left[\frac{1}{2Q_p} \pm \sqrt{\frac{1}{4Q_p^2} - 1}\right]$$

Real roots: $\frac{1}{4Q_p^2} - 1 \geq 0 \Rightarrow Q_p \leq \frac{1}{2}$

Complex roots: $s = -\omega_p' \left[\frac{1}{2Q_p} \pm j\sqrt{1 - \frac{1}{4Q_p^2}}\right]$; $Q_p > \frac{1}{2}$

$$s = -\omega_p' \left[\frac{1}{2Q_p} \pm \sqrt{\frac{1}{4Q_p^2} - 1}\right]$$

$$D(s) = \frac{s^2}{\omega_p'^2} + \frac{s}{\omega_p' Q_p} + 1$$

$ax^2 + bx + c = 0$

$x_1 + x_2 = \frac{-b}{a}$, $x_1 x_2 = c/a$

if $x_1 \gg x_2$

$\Rightarrow x_1 + x_2 \approx x_1 = \frac{-b}{a}$

$\downarrow$

$x_2 = \frac{-c}{b}$

$$s_1 + s_2 = -\frac{1/\omega_p' Q_p}{1/\omega_p'^2} = -\frac{\omega_p'}{Q_p}$$

$s_1 \gg s_2 \Rightarrow s_1 = -\omega_p' / Q_p$

$$s_1 s_2 = \omega_p'^2$$

$$s_2 = \frac{\omega_p'^2}{-\omega_p'/Q_p} = -\omega_p' Q_p$$

$s_1 = -\omega_p'/Q_p$ (high freq. pole) $= -\omega_{ph}$

$s_2 = -\omega_p' Q_p$ (Dominant pole) $= -\omega_{pl}$

$$\omega_p' = \omega_p \sqrt{\frac{A_{02}}{K} + 1}$$

$$H_{cl}(s) = \frac{K}{1 + \frac{K}{A_{02}}} \cdot \frac{1}{1 + s/\omega_{pl}} \times \frac{1}{1 + s/\omega_{ph}}$$

$$V_o(s) = V_i(s)\, H_{cl}(s)$$

$$= \frac{1}{s} \cdot \frac{K}{1 + \frac{K}{A_0}} \cdot \frac{1}{1 + s/\omega_{pl}} \times \frac{1}{1 + s/\omega_{ph}}$$

$v_o(t) = \mathcal{L}^{-1}\{\;\}$

$$A(s) = \frac{A_{02}}{\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + s/\omega_{p2}\right)}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{K}{1 + \frac{K}{A_{02}}\left(1 + \frac{s}{\omega_{p1}}\right)\left(1 + \frac{s}{\omega_{p2}}\right)}$$

$$= \frac{K}{1 + \frac{K}{A_{02}} + \frac{K}{A_{02}} s\left(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}}\right) + \frac{K}{A_{02}} \frac{s^2}{\omega_{p1}\omega_{p2}}}$$

$$= \frac{k \left/ \left(1 + k/A_{02}\right)\right.}{1 + \frac{1}{\left(\frac{A_{02}}{k}+1\right)} s\left(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}}\right) + \frac{1}{\left(\frac{A_{02}}{k}+1\right)} \frac{s^2}{\omega_{p1}\omega_{p2}}}$$

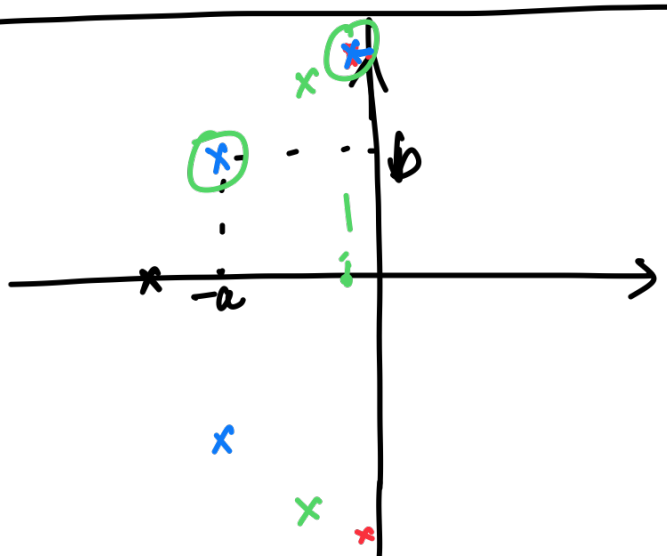
$$= \frac{}{1 + \frac{s}{\omega_p' Q_p} + \frac{s^2}{\omega_p'^2}}$$

$$\omega_p' = \sqrt{\left(\frac{A_{02}}{k}+1\right) \omega_{p1}\omega_{p2}}$$

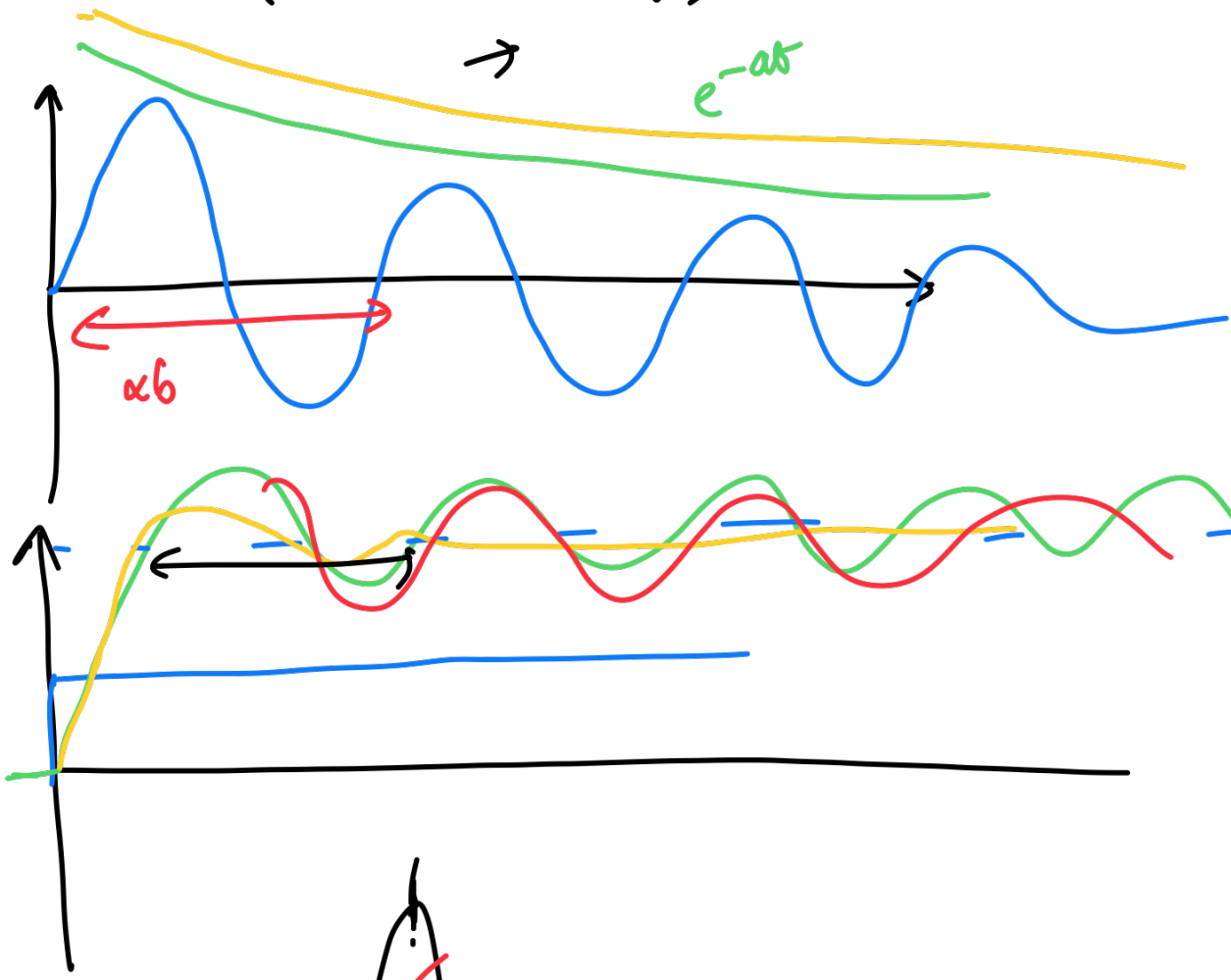
$$\omega_p' Q_p = \left(\frac{A_{02}}{k}+1\right) \frac{\omega_{p1}\omega_{p2}}{\omega_{p1}+\omega_{p2}}$$

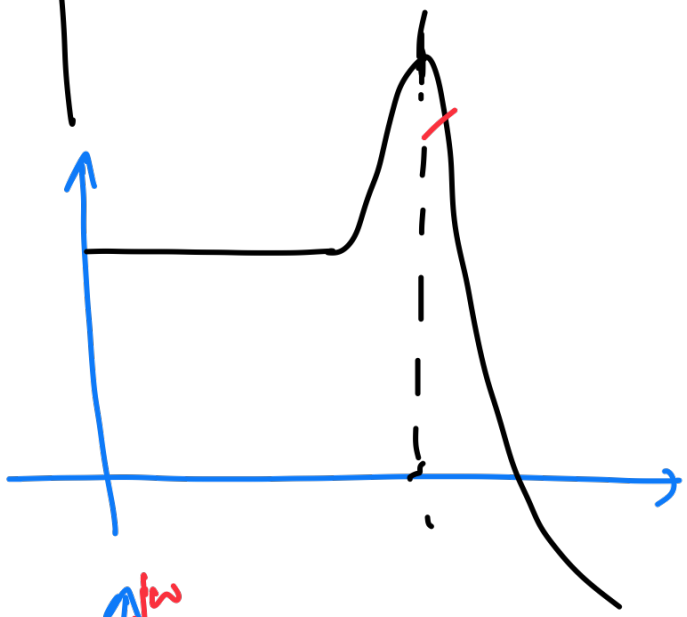
$$Q_p = \sqrt{\frac{A_{02}}{k}+1} \; \frac{\sqrt{\omega_{p1}\omega_{p2}}}{\omega_{p1}+\omega_{p2}}$$

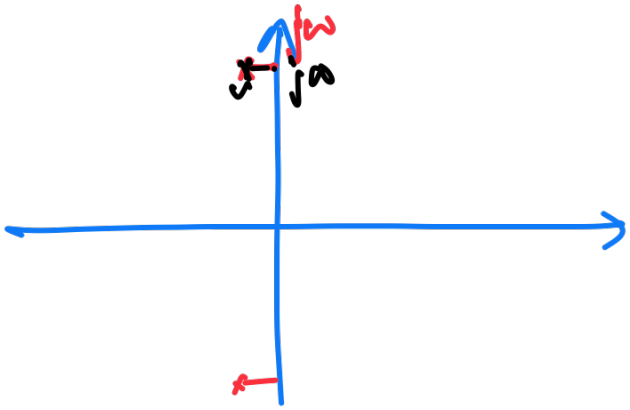
$$= \sqrt{\frac{A_{02}}{k}+1} \; \frac{1}{\sqrt{\frac{\omega_{p1}}{\omega_{p2}}} + \sqrt{\frac{\omega_{p2}}{\omega_{p1}}}}$$



$$H_{CL} = \frac{k'}{(s+a+jb)(s+a-jb)} \quad ; \quad a>0$$







$$H(j\omega) = \frac{\qquad}{|s+a+jb|\,(\quad)}$$

$s \to j\omega$